PREDICTING

Loan Default Status

Using Random Forest Classification

DATA SCIENCE CAPSTONE

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WGU D606 — Data Science Capstone | EBN2 Task 3

February 2026

About Me

PRESENTER

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Background & Focus Areas:

- Data Science & Machine Learning
- Financial Analytics & Risk Modeling
- Predictive Modeling & Classification

ABOUT THIS PRESENTATION

Today's Goal

Share how a machine learning model can help lenders predict which borrowers are most likely to default on their loans — in plain, everyday terms.

What We'll Cover

- The problem we set out to solve
- How we gathered and prepared the data
- What we found and what it means
- Practical recommendations for lenders

The Problem — Why This Matters

Every year, lenders lose billions of dollars when borrowers fail to repay their loans. The challenge? Identifying risky borrowers before a loan is approved.

The Risk

Loan defaults cause major financial losses for banks and credit unions, contributing to economic instability (CFPB, 2022).

The Gap

Traditional credit scoring misses complex patterns — like how a high debt load combined with a low income creates outsized risk.

The Opportunity

Machine learning can analyze hundreds of factors simultaneously, giving lenders a smarter early warning system.

Research Question: To what extent do financial characteristics — loan amount, income, credit score, and debt-to-income ratio — affect a borrower's likelihood of defaulting?

Statement of the Problem & Hypothesis

Null Hypothesis (H_o)

The model will NOT predict loan defaults better than a random guess — meaning borrower financial attributes have no meaningful relationship to default risk.

Alternate Hypothesis (H_1)

The model WILL predict loan defaults with accuracy and precision meaningfully above chance — confirming that financial attributes are effective predictors of default risk.

How We Defined 'Success'

Because this is a machine learning model (not a traditional statistics test), we measured success through real-world performance metrics rather than p-values:

Accuracy

What % of all loans did the model classify correctly?

Precision

Of loans flagged as defaults, how many actually defaulted?

Recall

Of all actual defaults, how many did the model catch?

ROC-AUC

Overall ability to separate defaulters from non-defaulters (0.5 = random, 1.0 = perfect)

The Data — What We Worked With

148,670

loan records

from WGU's
Loan Default Dataset

14 Key Variables Analyzed

- Gender
- Loan Limit
- Loan Purpose
- Business vs. Personal
- Region
- Age
- Credit Type
- Occupancy Type
- Loan Amount ($)
- Annual Income ($)
- Credit Score
- Debt-to-Income Ratio
- Property Value ($)
- **Default Status (Target)**

Class Imbalance — A Major Challenge:

Most loans in the dataset were repaid. Only ~16% of loans in our cleaned dataset resulted in defaults — making it harder for the model to learn what default looks like.



Summary of the Data Analysis Process

1 Import & Inspect

Loaded 148,670 loan records from Excel. Confirmed 34 columns and data types.

2 Select Features

Narrowed to 14 relevant columns: 8 categorical (demographics) + 5 financial + 1 target.

3 Clean the Data

Removed rows with missing values. Final dataset: 121,447 records (81.7% retained).

4 Encode Categories

Converted text categories (e.g., Region, Gender) into numerical format the model can read.

5 Split the Data

70% for training, 15% for validation, 15% for final testing — stratified to preserve class balance.

6 Train & Tune Model

Trained a Random Forest model, then optimized it using a grid search across 48 combinations.

Our Approach — The Random Forest Model

What Is a Random Forest?

Imagine asking 500 different financial experts to each review a loan application and vote on whether it will default. The final answer is whichever outcome the majority votes for.

That's essentially what a Random Forest does — it builds 500 independent decision trees and combines their votes for a final, more reliable prediction.

Why Random Forest?

✓ Handles Complexity

Detects non-obvious patterns that simple rules miss (e.g., high debt + low income together)

✓ Handles Imbalanced Data

We set class_weight='balanced' so the model pays extra attention to the rare default cases

✓ Prevents Overfitting

By averaging across 500 trees, it avoids memorizing the training data and generalizes better

✓ Provides Feature Importance

Shows us which financial factors matter most for predicting defaults

Outline of Findings — Final Model Results

After training on 85,012 loans and testing on 18,218 loans the model had never seen:

84.7%

Accuracy

Correctly classified 84.7% of all loans

73.7%

ROC-AUC

Far above the 50% baseline of random chance

26.5%

Default Recall

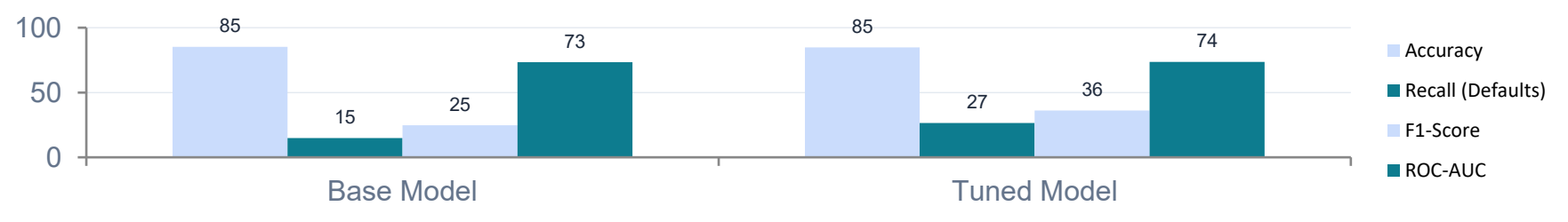
Correctly flagged 787 out of 2,973 defaults

57.0%

Precision

When it flagged a default, 57% were real defaults

Base Model vs. Tuned Model Comparison (%)



What the Model Learned — Key Insight

The model ranked defaulted loans above repaid loans more often than chance in the held-out test set.

Test Set Results — Who Did the Model Catch?

	Predicted: OK	Predicted: DEFAULT
Actual: OK	14,652 ✓ Correctly cleared	593 ✗ False alarms
Actual: DEFAULT	2,186 ✗ Missed defaults	787 ✓ Caught defaults

Held-out ranking

ROC-AUC 0.737

The score summarizes ranking across thresholds. At the selected threshold, the model caught 26.5% of defaults.

The biggest challenge: the model missed 2,186 defaults. Recall is the key metric to improve in future iterations (Lessmann et al., 2015).

Limitations of the Analysis

01

Single Dataset — Limited Generalizability

The model was trained and tested on one institutional dataset from one time period. Results may not apply to other lenders, regions, or economic conditions. External validation was not performed.

02

Missing Data Removed — Possible Selection Bias

We dropped 27,223 rows with missing values. If those borrowers had different characteristics than the rest, the model may not represent them fairly — a form of selection bias.

03

Low Recall — Many Defaults Still Missed

Even after tuning, the model only caught 26.5% of actual defaults. Nearly 3 out of every 4 defaulters in the test set were not flagged, limiting practical utility as a standalone tool.

04

Fairness & Bias Not Evaluated

The dataset contains sensitive variables (Gender, Region, Age). We did not measure whether the model's error rates differ across demographic groups — a critical ethical and regulatory gap (Barocas & Hardt, 2019).

05

Black Box — Hard to Explain Individual Decisions

With 500 decision trees, it is difficult to explain to a loan applicant exactly why they were flagged as high-risk. Tools like SHAP values would be needed to improve transparency.

Summary of Proposed Actions

Based on our findings, here is what we recommend for financial institutions:

Step 1

Deploy as a Decision-Support Tool

Use the model to flag high-risk applications for closer review — not to automatically deny loans. A human underwriter should make the final call on borderline cases.

Step 2

Calibrate the Risk Threshold

Adjust how aggressively the model flags defaults based on the institution's risk tolerance. Lower the threshold → catch more defaults (higher recall) but review more applications manually.

Step 3

Upgrade the Model

Compare against more powerful algorithms (XGBoost, LightGBM) which often outperform Random Forest on imbalanced data. Use precision-recall curves to find the optimal cut-off point.

Step 4

Add Explainability & Fairness Auditing

Implement SHAP values so loan officers can explain individual decisions. Audit error rates across Gender, Region, and Age groups to ensure equitable treatment (Barocas & Hardt, 2019).

Expected Benefits of This Study

What the held-out test can tell an underwriting team

787
Defaults Caught

Of 2,973 defaults in the test set, 787 were flagged. Financial savings would need to be measured in deployment.

73.7%
ROC-AUC Score

The test ROC-AUC was 0.737. A score of 0.5 corresponds to chance ranking. This measures ranking across thresholds.

~81%
Data Preserved

Cleaning retained 121,447 complete records from 148,670. Dropping rows can change which borrowers the model represents.

Broader Impact

The score could help prioritize manual review. Validation at another lender and checks for subgroup performance would be needed before making underwriting decisions.

References

All sources are cited in APA format. In-text citations appear throughout the presentation where content is paraphrased or referenced.

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Thank You

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Key Takeaway: A Random Forest model can meaningfully predict loan defaults (ROC-AUC 0.737), offering lenders a powerful early warning tool to reduce financial risk.